

Research-oriented compiler engineer who turns ambiguous compiler/language-system problems into explicit models, prototypes them, records applicability boundaries, and tests them with counterexamples and reproducible experiments. Professional grounding comes from LLVM work at MCST and industrial static analysis at ISP RAS; the strongest independent evidence is UniversalToolchain/contract-guided reverification and a conservative interprocedural Global-IV pass.

EDUCATION

HSE University — Software Engineering, 2026–2030

Moscow / remote

EXPERIENCE

- 2026 — present

ISP RAS — Static Analysis Engineer

SharpChecker · C#.NET

• Contribute to SharpChecker, ISP RAS's industrial static-analysis platform for C#.NET.
- July — August 2026

MCST — Compiler Engineering Intern

LLVM 22 · C++23 · compiler engineering

• Implemented an LLVM LICM pass with loop analysis, side-effect/speculative-safety checks, and hoisting of loop-invariant instructions to preheaders.

• Developed a conservative interprocedural global-IV prototype with APInt affine evolution, transitive call effects, a separate legality phase, and LLVM IR transformation; unsupported CFG/call cases are rejected.

SELECTED RESEARCH & ENGINEERING

- UniversalToolchain

Evidence-backed anonymous CGO'27 draft, not a submission: four policies, a frozen corpus, valid controls, fault injection, and ablations; performance/external-corpus claims remain explicitly blocked.
- LLVM Interprocedural Global IV Optimization

29 transform/reject regressions; LLVM Verifier, second-pass idempotence, and executable before/after comparison act as independent correctness oracles.
- x86-64 Codegen & Register Allocation Lab

A reference interpreter and differential execution check generated-code semantics; the allocator contract is verified independently.
- DerefAfterNullAnalyzer

Successors are reprocessed until the state stabilizes, producing diagnostics for potentially unsafe dereference paths.

SKILLS

Research Engineering	hypothesis-driven prototyping, controlled experiments, frozen corpora, fault injection, matched controls, ablations, explicit validity boundaries
Compilers / Program Analysis	LLVM 22, LLVM IR, CFG/SSA, dominators, loop/interprocedural analysis, Roslyn CFG, fixed-point data flow, transformation legality
Language Systems	typed artifact contracts, feature/component composition, deterministic planning, provider/order/conflict resolution, exact runtime binding
Verification & Backends	LLVM Verifier, idempotence checks, differential execution, interpreter/CIL parity, x86-64 codegen, register-allocation verification

RECOGNITION

- LangDev'26 — Build the Language, Then Make the Abstractions Disappear

Accepted talk on UniversalToolchain's architecture; LangDev'26 takes place October 8–9, 2026 in Málaga.
- UniversalToolchain — Baltic Science and Engineering Competition

First Degree Diploma and Grand Prize "Perfection as Hope" for UniversalToolchain.